

Brief summary of this function: Plots the transition points on the graphs containing the plotcycle_modes

Detailed explanation of this function: This function plots the transition points on the graphs when it switches phases with yellow star shaped dots. This is made to find the points where there is a change in charging mode, store it in the array, then plot those points when they are there.

```
function plotTransition_points(time,y,mode) %naming the helper function

%finding the change in the charging mode
transitionIdx = []; %creating an empty array

for i = 1:length(mode)-1 %start of the loop

    %this checks that the current charging mode is not the same as the next
    %mode
    if mode(i) ~= mode(i+1)
        transitionIdx(end+1) = i; %adds current index to end of the array
    end
end

%plotting the points on the graph for when there is a transition

if isempty(transitionIdx) == 0 %makes sure that the array is empty
    hold on
    %plot for the transition points (appears as a yellow star shape on
    %graph)
    plot(time(transitionIdx), y(transitionIdx), '*', 'MarkerFaceColor','y',
'MarkerSize', 4, ...
        'Color', 'y', 'DisplayName','Transition Point')
    hold off
end

end
```